

ROTEIRO DO RELATÓRIO TÉCNICO: MONITORAMENTO COM NAGIOS (Diego Machado,Magno Guimaraes)

1. Infraestrutura e Rede (AWS)

Nesta etapa, foram provisionadas as instâncias na nuvem AWS e configuradas as regras de firewall para permitir a comunicação entre o servidor de monitoramento e os clientes.

☰

EC2

>

Instâncias

EC2

<

Painel

Visualização Global do EC2

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Eventos

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AMIs

Catálogo de AMIs

▼ Elastic Block Store

Volumes

Snapshots

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Paras de chaves

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Localizar Instância por atributo ou tag (case-sensitive)

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< 1 >

⚙️

☑

Name 🔗 ▼

ID da instância

Estado da inst... ▼

Tipo de inst... ▼

Verificação de stat

Status

☒

windows nagios

i-051150f39862d8f57

✅ Executando 🔍 🔍

t3.large

✅ 3/3 verificações a

Exibir

☐

maquina 3

i-08c0f4d404fb358f

✅ Executando 🔍 🔍

t3.micro

✅ 3/3 verificações a

Exibir

☐

nagios

i-03f44cfb5a9f15f0e

✅ Executando 🔍 🔍

t2.nano

✅ 2/2 verificações a

Exibir

☐

maquina 4

i-0ceabaa2f3b12564c

✅ Executando 🔍 🔍

t3.micro

🕒 Inicializando

Exibir

i-051150f39862d8f57 (windows nagios)

⚙️

▼

Grupos de segurança

🔗

sg-0d57a6b5e04f1940a (launch-wizard-1)

▼ Regras de entrada

🔍

Regras de filtro

Legenda: Figura 1: Console AWS EC2 exibindo as quatro instâncias provisionadas (1 Servidor Ubuntu, 2 Clientes Linux, 1 Cliente Windows) em estado de execução.

Catálogo de AMIs

▼ Elastic Block Store

Volumes

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Security groups

IPs elásticos

i-051150f39862d8f57 (windows nagios)

–	sgr-094d54886891a5f0e	3306	TCP	0.0.0.0/0
–	sgr-0c826133b55a5975e	443	TCP	0.0.0.0/0
–	sgr-03fce83a1ed6c4986	22	TCP	0.0.0.0/0
–	sgr-02538f7845ec2d4e3	Todos	Todos	0.0.0.0/0
–	sgr-0cc47b71f8af036e8	5666	TCP	0.0.0.0/0
–	sgr-0cf27bf5f3f61d1aa	80	TCP	0.0.0.0/0

Legenda: Figura 2: Configuração do Grupo de Segurança (Firewall). Destaque para a liberação das portas TCP 5666 (Agente NRPE Linux) e TCP 12489/3389 (Agente Windows e RDP), garantindo a conectividade do monitoramento.

2. Instalação do Servidor Nagios (Ubuntu)

Foi realizada a compilação manual do Nagios Core e dos Plugins de monitoramento no servidor Linux principal.

```
ubuntu@ip-172-31-16-168:~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1305 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1585 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [299 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.7 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1498 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [303 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [215 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.4 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2244 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9336 B]
```

Legenda: Figura 3: Preparação do ambiente Linux com atualização de repositórios (*apt update*) e instalação de dependências essenciais de compilação.

```
ubuntu@ip-172-31-16-168:~$ cd /tmp
wget https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.2.tar.gz
tar xzf nagios-4.5.2.tar.gz
cd nagios-4.5.2
sudo ./configure --with-httpd-conf=/etc/apache2/sites-enabled
sudo make all
--2025-11-17 23:41:16-- https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.2.tar.gz
Resolving assets.nagios.com (assets.nagios.com)... 45.79.49.120, 2600:3c00::f03c:92ff:fe7f:45ce
Connecting to assets.nagios.com (assets.nagios.com)|45.79.49.120|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11540257 (11M) [application/x-gzip]
Saving to: 'nagios-4.5.2.tar.gz'

nagios-4.5.2.tar.gz      100%[=====] 11.00M  10.4MB/s  in 1.1
```

Legenda: Figura 4: Download do código-fonte do Nagios Core 4.5.2 e processo de compilação (*make all*) no servidor.

```
ubuntu@ip-172-31-16-168:~/nagioscore-nagios-4.4.14$ sudo make install-config
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc/objects
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/nagios.cfg /usr/local/nagios/etc/nagios.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg /usr/local/nagios/etc/cgi.cfg
/usr/bin/install -c -b -m 660 -o nagios -g nagios sample-config/resource.cfg /usr/local/nagios/etc/resource.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/templates.cfg /usr/local/nagios/etc/obj
ts/templates.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/commands.cfg /usr/local/nagios/etc/obj
s/commands.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/contacts.cfg /usr/local/nagios/etc/obj
s/contacts.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/timeperiods.cfg /usr/local/nagios/etc/
ects/timeperiods.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/localhost.cfg /usr/local/nagios/etc/ob
ts/localhost.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/windows.cfg /usr/local/nagios/etc/obje
/windows.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/printer.cfg /usr/local/nagios/etc/obje
/printer.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/switch.cfg /usr/local/nagios/etc/objec
switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

ubuntu@ip-172-31-16-168:~/nagioscore-nagios-4.4.14$
```

Legenda: Figura 5: Instalação dos binários, scripts de inicialização e arquivos de configuração base do Nagios.

```
ts/localhost.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/windows.cfg /usr/local/nagios/etc/o
/windows.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/printer.cfg /usr/local/nagios/etc/o
/printer.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/switch.cfg /usr/local/nagios/etc/ob
switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/sites-enabled/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/apache2/sites-enabled/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

Module rewrite already enabled
Module cgi already enabled
ubuntu@ip-172-31-16-168:/tmp/nagios-4.5.2$ sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
```

Legenda: Figura 6: Configuração da autenticação web via `htpasswd`, criando o usuário 'nagiosadmin' para acesso seguro ao painel.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-16-168:/tmp/nagios-4.5.2$ sudo systemctl restart apache2
sudo systemctl start nagios
sudo systemctl enable nagios
ubuntu@ip-172-31-16-168:/tmp/nagios-4.5.2$ systemctl status nagios
● nagios.service - Nagios Core 4.5.2
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-17 23:47:32 UTC; 3min 37s ago
     Docs: https://www.nagios.org/documentation
    Main PID: 9318 (nagios)
      Tasks: 6 (limit: 517)
     Memory: 7.7M (peak: 10.1M)
        CPU: 91ms
    CGroup: /system.slice/nagios.service
            └─9318 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
              └─9319 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                └─9320 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                  └─9321 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                    └─9322 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                      └─9327 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: qh: Socket '/usr/local/nagios/var/rw/nagios.qh' successfully initia
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: qh: core query handler registered
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: qh: echo service query handler registered
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: qh: help for the query handler registered
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: wproc: Successfully registered manager as @wproc with query handler
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: wproc: Registry request: name=Core Worker 9319;pid=9319
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: wproc: Registry request: name=Core Worker 9321;pid=9321
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: wproc: Registry request: name=Core Worker 9322;pid=9322
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: wproc: Registry request: name=Core Worker 9320;pid=9320
Nov 17 23:47:32 ip-172-31-16-168 nagios[9318]: Successfully launched command file worker with pid 9327
ubuntu@ip-172-31-16-168:/tmp/nagios-4.5.2$
```

Legenda: Figura 7: Verificação final do serviço Nagios (Systemd), confirmando status 'Active (Running)' após a instalação.

3. Configuração dos Agentes Linux (NRPE)

Nos clientes Linux, foi instalado o agente NRPE para permitir que o servidor execute verificações locais de CPU e Memória.

```
admin@ip-172-31-72-36:~$ sudo apt update
sudo apt install -y nagios-nrpe-server nagios-plugins
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [47 B]
Get:3 https://cdn-aws.deb.debian.org/debian trixie InRelease [140 kB]
Get:4 https://cdn-aws.deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:5 https://cdn-aws.deb.debian.org/debian trixie-backports InRelease [54.0 kB]
Get:6 https://cdn-aws.deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:7 https://cdn-aws.deb.debian.org/debian trixie/main Sources [10.5 MB]
Get:8 https://cdn-aws.deb.debian.org/debian trixie/main amd64 Packages [9670 kB]
Get:9 https://cdn-aws.deb.debian.org/debian trixie/main Translation-en [6484 kB]
Get:10 https://cdn-aws.deb.debian.org/debian trixie-updates/main Sources [2788 B]
Get:11 https://cdn-aws.deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:12 https://cdn-aws.deb.debian.org/debian trixie-updates/main Translation-en [4096 B]
Get:13 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources [86.1 kB]
Get:14 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Packages [111 kB]
Get:15 https://cdn-aws.deb.debian.org/debian trixie-backports/main Translation-en [85.7 kB]
Get:16 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Sources [93.2 kB]
Get:17 https://cdn-aws.deb.debian.org/debian-security trixie-security/main amd64 Packages [71.8 kB]
Get:18 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Translation-en [46.9 kB]
Fetched 27.5 MB in 2s (11.3 MB/s)
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
Note, selecting 'monitoring-plugins' instead of 'nagios-plugins'
Upgrading:
  bind9-host  bind9-libs

Installing:
  monitoring-plugins  nagios-nrpe-server

Installing dependencies:
  bind9-dnsutils  libpython3.13  mysql-common
  libarchive13t64  libradscli4    perl
```

Legenda: Figura 8: Instalação do servidor NRPE e plugins de monitoramento nas instâncias clientes Linux (Debian/Ubuntu).

```
GNU nano 8.4 /etc/nagios/nrpe.cfg *
# This is an optional comma-delimited list of IP address or hostnames
# that are allowed to talk to the NRPE daemon. Network addresses with a bit mask
# (i.e. 192.168.1.0/24) are also supported. Hostname wildcards are not currently
# supported.
#
# Note: The daemon only does rudimentary checking of the client's IP
# address. I would highly recommend adding entries in your /etc/hosts.allow
# file to allow only the specified host to connect to the port
# you are running this daemon on.
#
# NOTE: This option is ignored if NRPE is running under either inetd or xinetd

allowed_hosts=127.0.0.1,::1,34.203.234.166

# COMMAND ARGUMENT PROCESSING
# This option determines whether or not the NRPE daemon will allow clients
# to specify arguments to commands that are executed. This option only works
# if the daemon was configured with the --enable-command-args configure script
# option.
#
# *** ENABLING THIS OPTION IS A SECURITY RISK! ***
# Read the SECURITY file for information on some of the security implications
# of enabling this variable.
#
# Values: 0=do not allow arguments, 1=allow command arguments

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Ma
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/_ Go To Line M-E Redo     M-6 Copy
```

Legenda: Figura 9: Configuração de segurança no arquivo `nrpe.cfg` do cliente, autorizando explicitamente o IP Privado do Servidor Nagios a solicitar dados.

```
admin@ip-172-31-72-36: ~
Nov 19 01:33:20 ip-172-31-72-36 nrpe[4821]: Starting up daemon
Nov 19 01:33:20 ip-172-31-72-36 nrpe[4821]: Server listening on 0.0.0.0 port 5666.
Nov 19 01:33:20 ip-172-31-72-36 nrpe[4821]: Server listening on :: port 5666.
Nov 19 01:33:20 ip-172-31-72-36 nrpe[4821]: Listening for connections on port 5666
Nov 19 01:33:20 ip-172-31-72-36 nrpe[4821]: Allowing connections from: 127.0.0.1,::1,34.203.234.166

[1]+  Stopped                  systemctl status nagios-nrpe-server
admin@ip-172-31-72-36:~$ sudo nano /etc/nagios/nrpe.cfg
admin@ip-172-31-72-36:~$ sudo systemctl restart nagios-nrpe-server
admin@ip-172-31-72-36:~$ sudo systemctl restart nagios-nrpe-server
admin@ip-172-31-72-36:~$ systemctl status nagios-nrpe-server
● nagios-nrpe-server.service - Nagios Remote Plugin Executor
   Loaded: loaded (/usr/lib/systemd/system/nagios-nrpe-server.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-11-19 01:46:34 UTC; 8s ago
     Invocation: e2a882aef5ef46ddb104d2bbb12c887e
       Docs: http://www.nagios.org/documentation
    Main PID: 4856 (nrpe)
      Tasks: 1 (limit: 1097)
     Memory: 1.4M (peak: 1.8M)
        CPU: 16ms
       CGroup: /system.slice/nagios-nrpe-server.service
              └─4856 /usr/sbin/nrpe -c /etc/nagios/nrpe.cfg -f

Nov 19 01:46:34 ip-172-31-72-36 systemd[1]: Started nagios-nrpe-server.service - Nagios Remote Plugin Executor.
Nov 19 01:46:34 ip-172-31-72-36 (nrpe)[4856]: nagios-nrpe-server.service: Referenced but unset environment variable eval
Nov 19 01:46:34 ip-172-31-72-36 nrpe[4856]: Starting up daemon
Nov 19 01:46:34 ip-172-31-72-36 nrpe[4856]: Server listening on 0.0.0.0 port 5666.
Nov 19 01:46:34 ip-172-31-72-36 nrpe[4856]: Server listening on :: port 5666.
Nov 19 01:46:34 ip-172-31-72-36 nrpe[4856]: Listening for connections on port 5666
Nov 19 01:46:34 ip-172-31-72-36 nrpe[4856]: Allowing connections from: 127.0.0.1,::1,172.31.16.168
lines 1-19/19 (END)
```

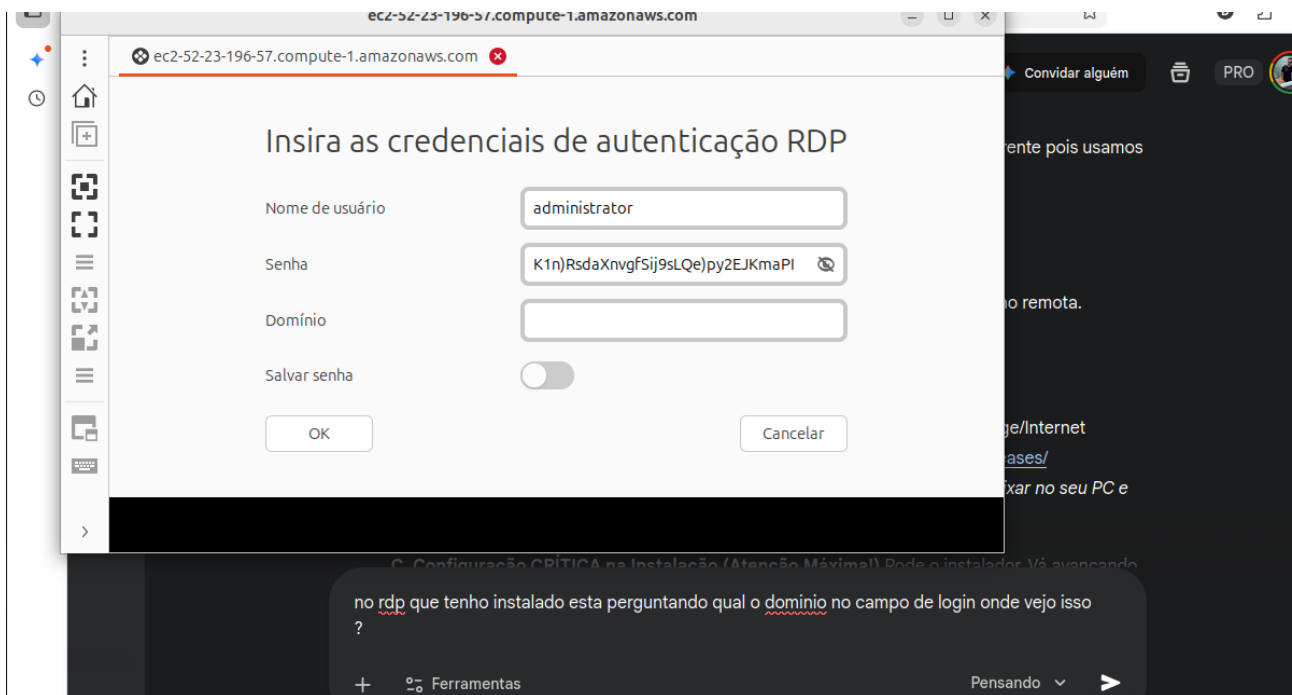
Legenda: Figura 10: Validação do serviço NRPE ativo no cliente Linux, pronto para receber conexões na porta 5666.


```
admin@ip-172-31-72-36: ~
ubuntu@ip-172-31-68-150: ~
ubuntu@ip-172-31-16-168: ~
ubuntu@ip-172-31-16-168:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.72.36
NRPE v4.1.3
ubuntu@ip-172-31-16-168:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.68.150
NRPE v4.1.0
ubuntu@ip-172-31-16-168:~$
```

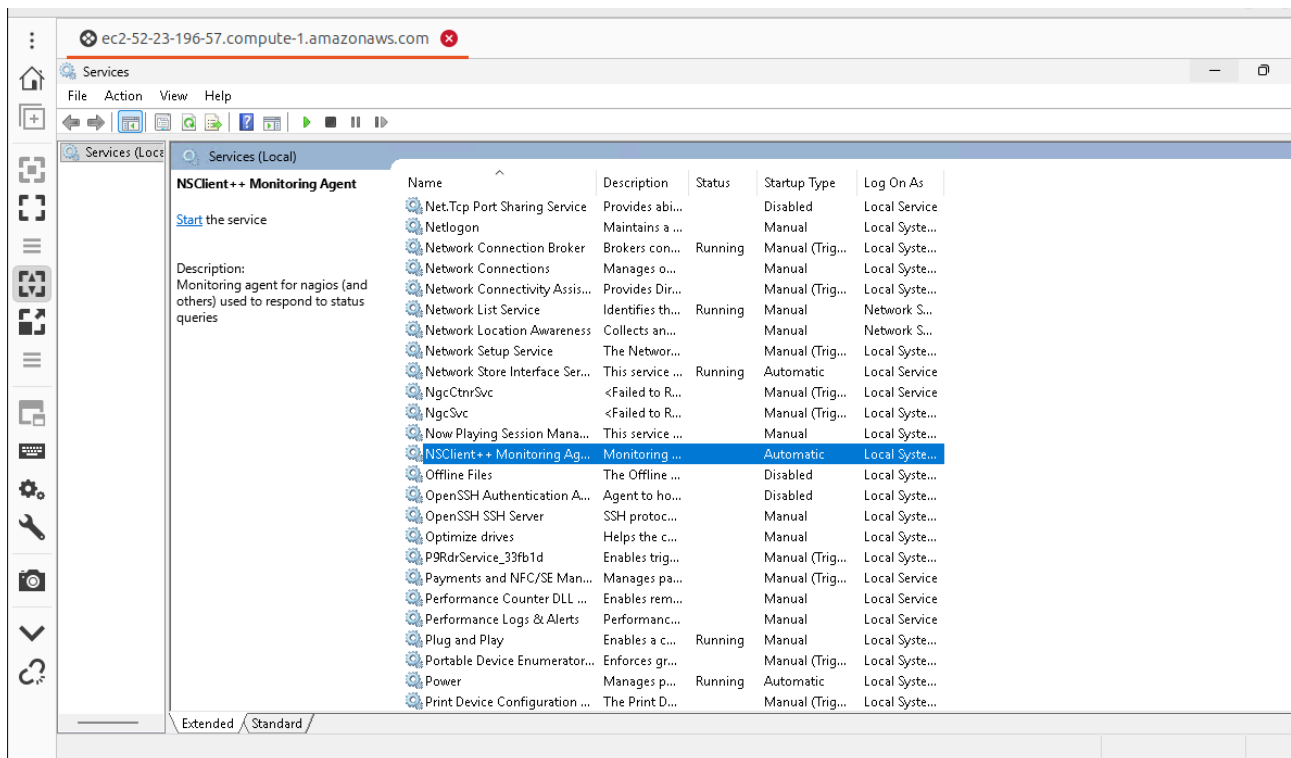
Legenda: Figura 11: Teste de conectividade realizado a partir do Servidor Nagios, confirmando comunicação bem-sucedida com os clientes via protocolo NRPE (retorno da versão v4.1.x).

4. Configuração do Agente Windows (NSClient++)

O host Windows foi configurado através de acesso remoto para instalação do agente NSClient++.



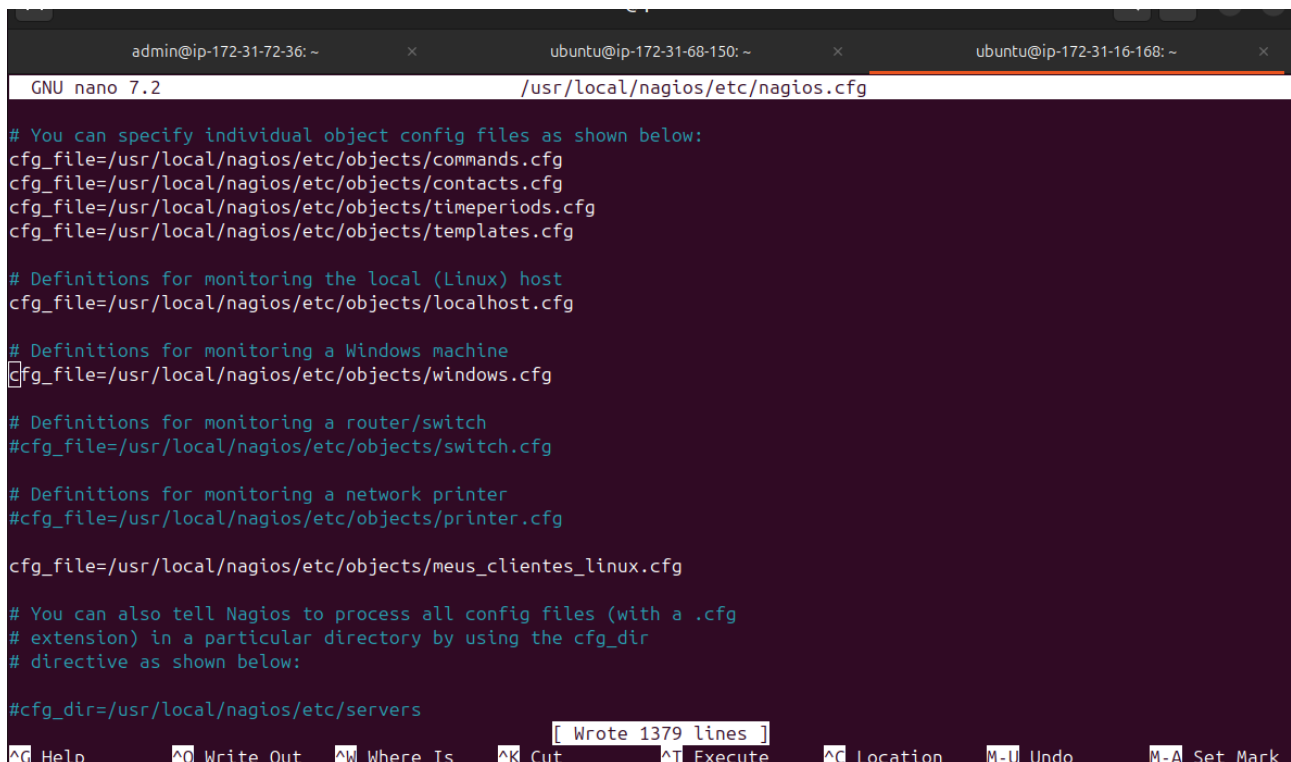
Legenda: Figura 12: Acesso à instância Windows via Remote Desktop Protocol (RDP) para administração e instalação do agente.



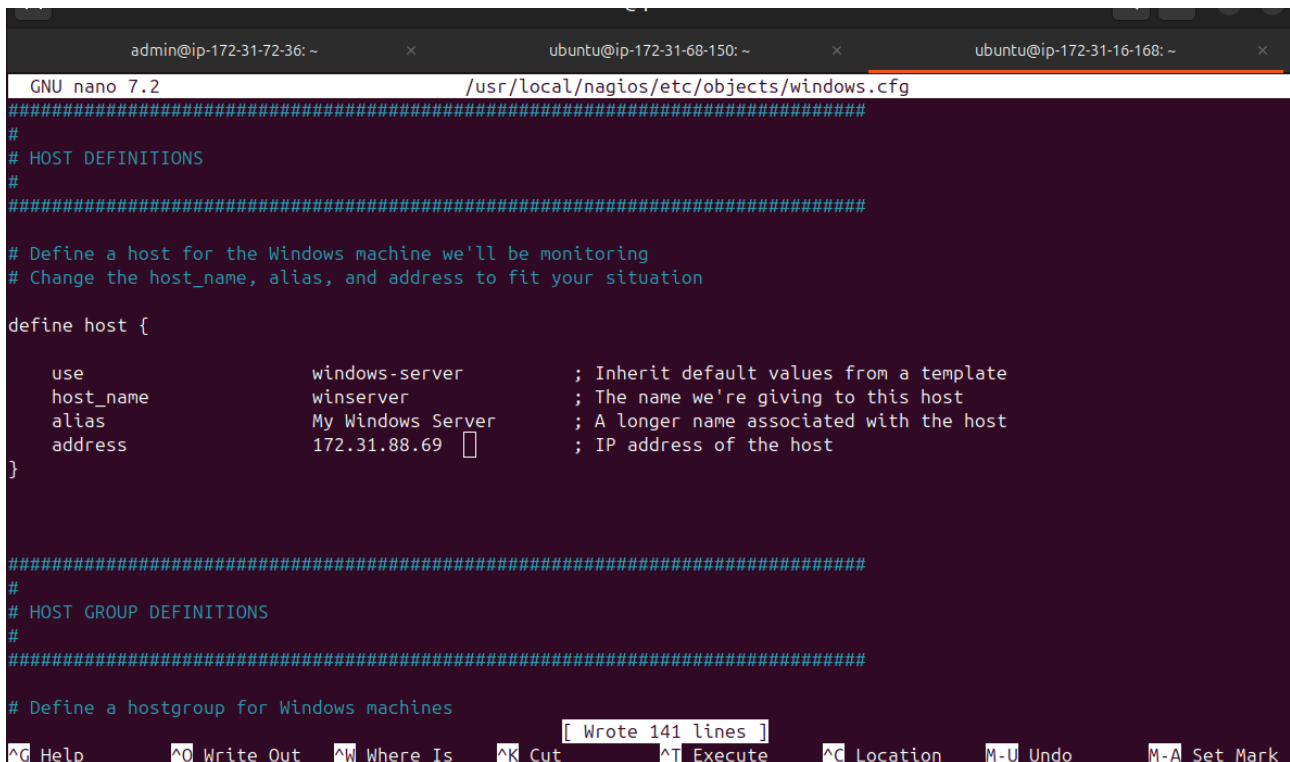
Legenda: Figura 13: Console de Serviços do Windows mostrando o agente 'NSClient++ Monitoring Agent' em execução e configurado para inicialização automática.

5. Integração e Configuração Final

O servidor Nagios foi configurado para reconhecer os novos hosts e comunicar-se com o Windows usando senha segura.



Legenda: Figura 14: Edição do arquivo principal `nagios.cfg`, habilitando a leitura das configurações para hosts Windows.



The screenshot shows a terminal window with three tabs: 'admin@ip-172-31-72-36: ~', 'ubuntu@ip-172-31-68-150: ~', and 'ubuntu@ip-172-31-16-168: ~'. The active tab is the third one, showing the 'GNU nano 7.2' editor editing '/usr/local/nagios/etc/objects/windows.cfg'. The file content includes comments for host definitions and a 'define host {' block with fields: 'use windows-server', 'host_name winserver', 'alias My Windows Server', and 'address 172.31.88.69'. A status bar at the bottom indicates '[Wrote 141 lines]' and shows various keyboard shortcuts like '^G Help', '^O Write Out', etc.

```
admin@ip-172-31-72-36: ~
ubuntu@ip-172-31-68-150: ~
ubuntu@ip-172-31-16-168: ~
GNU nano 7.2 /usr/local/nagios/etc/objects/windows.cfg
#####
#
# HOST DEFINITIONS
#
#####

# Define a host for the Windows machine we'll be monitoring
# Change the host_name, alias, and address to fit your situation

define host {

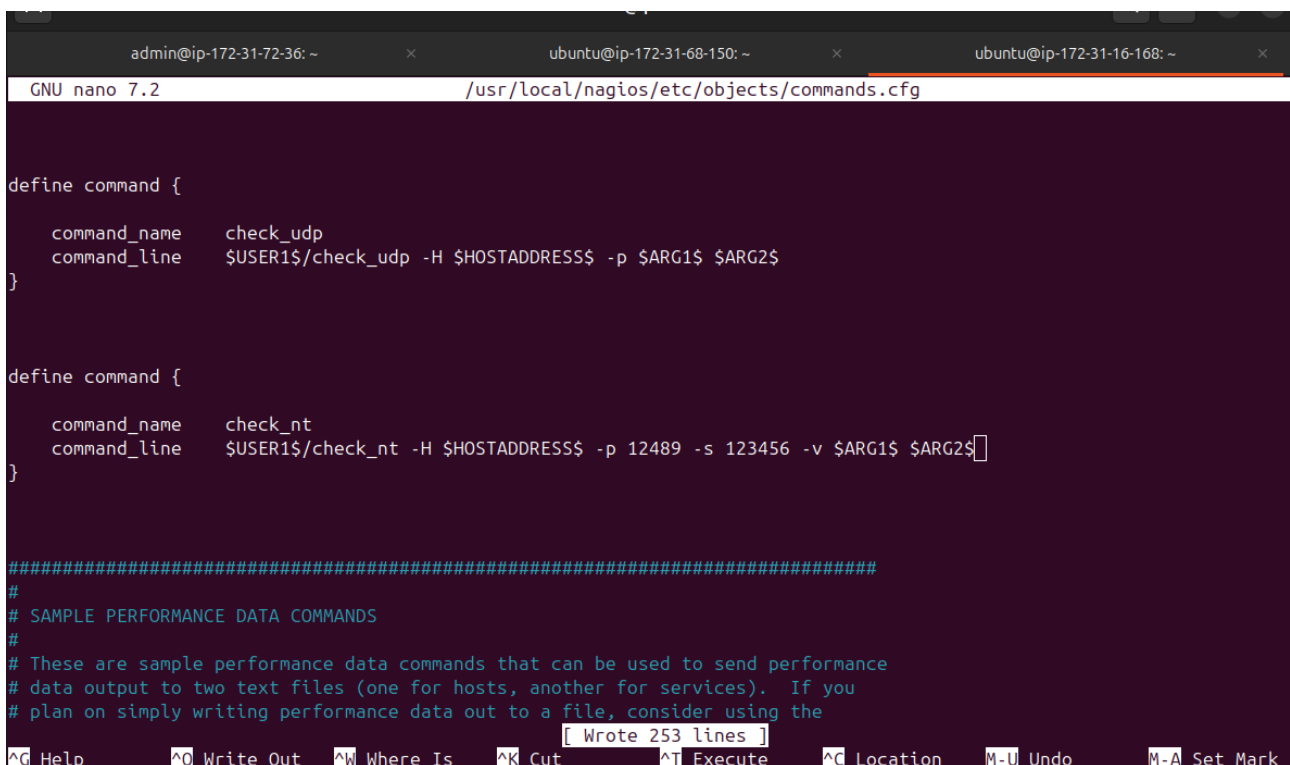
    use                windows-server        ; Inherit default values from a template
    host_name          winserver             ; The name we're giving to this host
    alias              My Windows Server     ; A longer name associated with the host
    address            172.31.88.69          ; IP address of the host
}

#####
#
# HOST GROUP DEFINITIONS
#
#####

# Define a hostgroup for Windows machines

[ Wrote 141 lines ]
^G Help  ^O Write Out  ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo    M-A Set Mark
```

Legenda: Figura 15: Definição do objeto Host Windows, apontando para o IP Privado da instância na AWS.



The screenshot shows the same terminal window with the 'GNU nano 7.2' editor editing '/usr/local/nagios/etc/objects/commands.cfg'. The file content includes two 'define command {' blocks. The first defines 'check_udp' with command_line '\$USER1\$/check_udp -H \$HOSTADDRESS\$ -p \$ARG1\$ \$ARG2\$'. The second defines 'check_nt' with command_line '\$USER1\$/check_nt -H \$HOSTADDRESS\$ -p 12489 -s 123456 -v \$ARG1\$ \$ARG2\$'. A status bar at the bottom indicates '[Wrote 253 lines]' and shows the same keyboard shortcuts as the previous image.

```
admin@ip-172-31-72-36: ~
ubuntu@ip-172-31-68-150: ~
ubuntu@ip-172-31-16-168: ~
GNU nano 7.2 /usr/local/nagios/etc/objects/commands.cfg

define command {

    command_name      check_udp
    command_line       $USER1$/check_udp -H $HOSTADDRESS$ -p $ARG1$ $ARG2$
}

define command {

    command_name      check_nt
    command_line       $USER1$/check_nt -H $HOSTADDRESS$ -p 12489 -s 123456 -v $ARG1$ $ARG2$
}

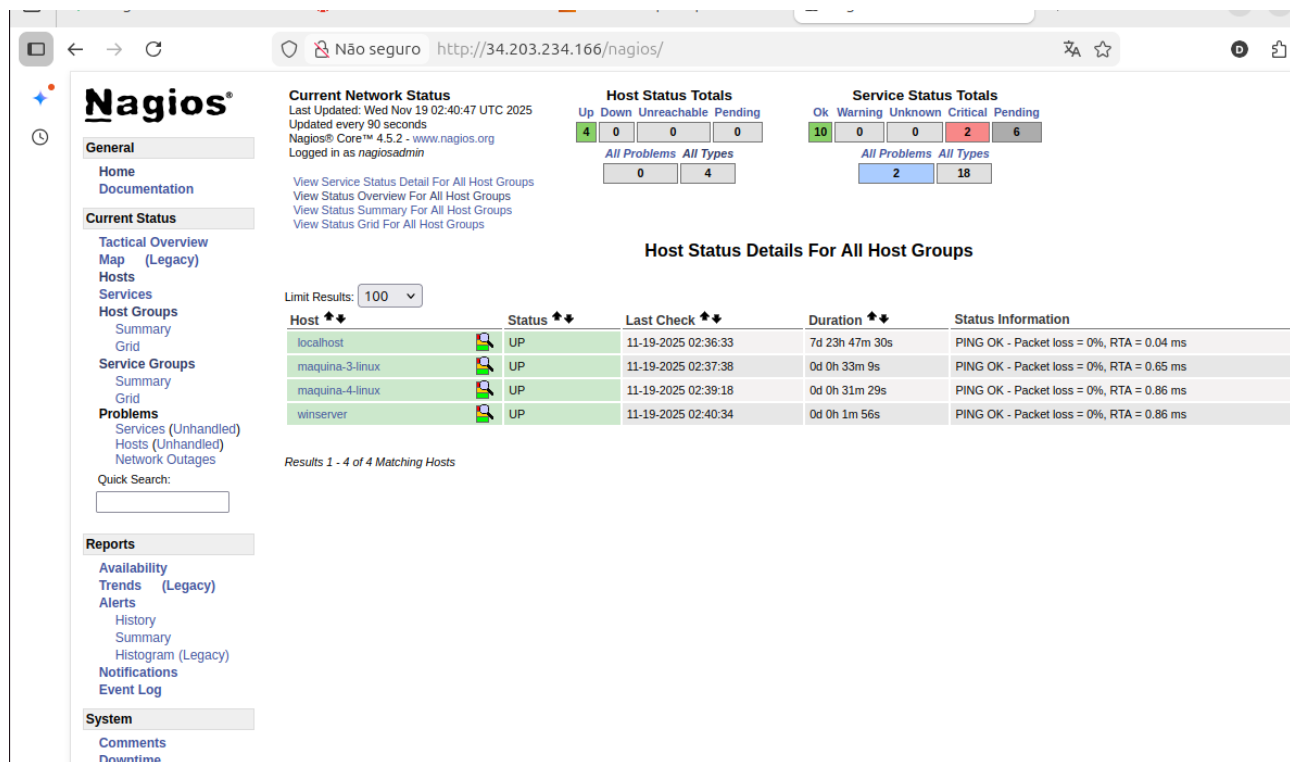
#####
#
# SAMPLE PERFORMANCE DATA COMMANDS
#
# These are sample performance data commands that can be used to send performance
# data output to two text files (one for hosts, another for services). If you
# plan on simply writing performance data out to a file, consider using the

[ Wrote 253 lines ]
^G Help  ^O Write Out  ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo    M-A Set Mark
```

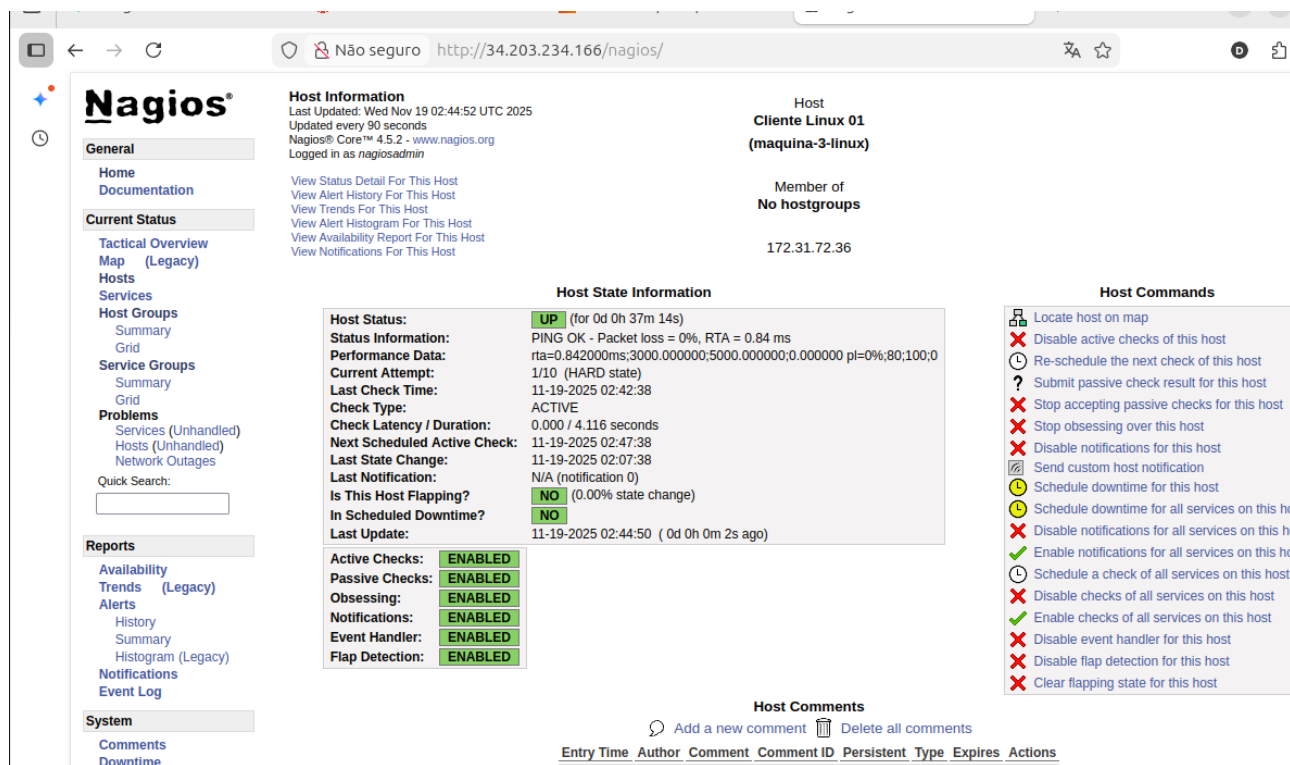
Legenda: Figura 16: Configuração de segurança no comando `check_nt`, inserindo a senha de autenticação para comunicação criptografada com o agente Windows.

6. Resultados Finais de Monitoramento

Apresentação do painel de controle operando em tempo real com todos os hosts monitorados.



Legenda: Figura 17: Dashboard Geral (Tactical Overview) exibindo 100% de disponibilidade. Servidor Local, 2 Clientes Linux e 1 Cliente Windows com status 'UP' (Verde).



Legenda: Figura 18: Detalhamento do host 'Cliente Linux debian', demonstrando a coleta de métricas de disponibilidade e tempo de resposta.

The screenshot displays the Nagios web interface for a host named 'My Windows Server (winserver)'. The interface is divided into several sections:

- General:** Includes links for Home, Documentation, and a list of navigation options like Tactical Overview, Map (Legacy), Hosts, Services, Host Groups, Service Groups, Problems, Reports, and System.
- Host Information:** Shows the host's last update time (Wed Nov 19 02:43:11 UTC 2025), update frequency (every 90 seconds), and login information (as nagiosadmin).
- Host State Information:** Displays the host's current status as 'UP' (for 0d 0h 4m 20s). It also shows performance data (PING OK, Packet loss = 0%, RTA = 0.93 ms), current attempt (1/10), last check time (11-19-2025 02:42:44), check type (ACTIVE), check latency/duration (0.000 / 4.064 seconds), next scheduled active check (11-19-2025 02:47:44), last state change (11-19-2025 02:38:51), last notification (N/A), and whether the host is flapping (NO).
- Host Commands:** A list of commands that can be executed on the host, such as 'Locate host on map', 'Disable active checks of this host', 'Re-schedule the next check of this host', etc.
- Host Comments:** A section for adding or deleting comments about the host.

Legenda: Figura 19: Detalhamento do host 'Windows Server', confirmando a integração bem-sucedida via NSClient++.

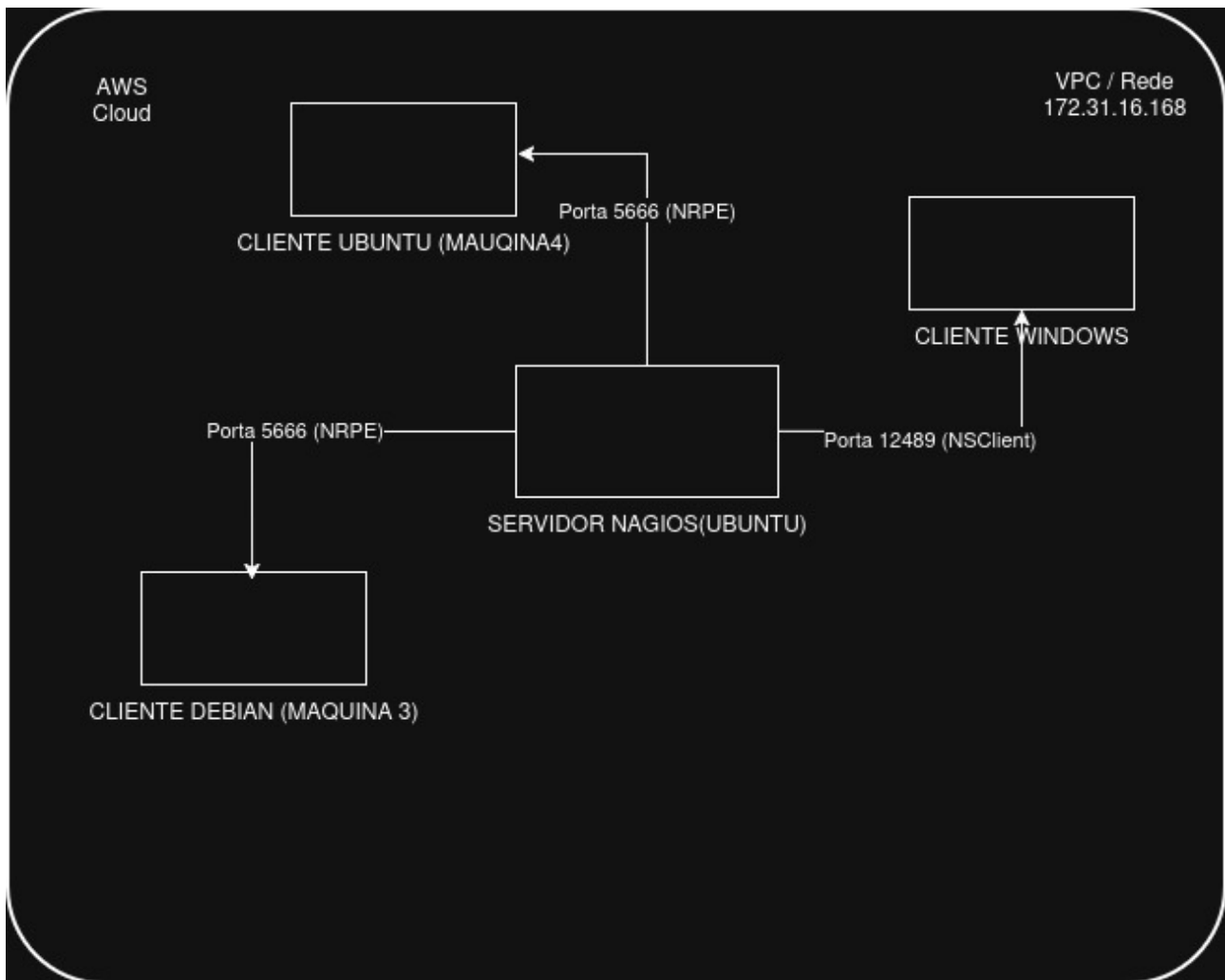
The screenshot displays the Nagios web interface for a host named 'Cliente Linux 02 (maquina-4-linux)'. The interface is divided into several sections:

- General:** Includes links for Home, Documentation, and a list of navigation options like Tactical Overview, Map (Legacy), Hosts, Services, Host Groups, Service Groups, Problems, Reports, and System.
- Host Information:** Shows the host's last update time (Wed Nov 19 02:45:29 UTC 2025), update frequency (every 90 seconds), and login information (as nagiosadmin).
- Host State Information:** Displays the host's current status as 'UP' (for 0d 0h 36m 11s). It also shows performance data (PING OK, Packet loss = 0%, RTA = 0.69 ms), current attempt (1/10), last check time (11-19-2025 02:44:18), check type (ACTIVE), check latency/duration (0.001 / 4.135 seconds), next scheduled active check (11-19-2025 02:49:18), last state change (11-19-2025 02:09:18), last notification (N/A), and whether the host is flapping (NO).
- Host Commands:** A list of commands that can be executed on the host, such as 'Locate host on map', 'Disable active checks of this host', 'Re-schedule the next check of this host', etc.
- Host Comments:** A section for adding or deleting comments about the host.

Legenda: Figura 20: Detalhamento do host 'Cliente Linux ubuntu', demonstrando a coleta de métricas de disponibilidade e tempo de resposta.

7. Topologia do Ambiente

Representação gráfica da arquitetura implementada.



Legenda: Figura 20: Diagrama de Topologia da solução, ilustrando a comunicação centralizada do Servidor Nagios com os agentes via portas 5666 (Linux) e 12489 (Windows) dentro da VPC AWS.